Question:
A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?
Answer Choices:
A. Autoimmune hemolytic anemia
B. Microangiopathic hemolytic anemia
C. Paroxysmal nocturnal hemoglobinuria (PNH)
D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<tool_call> reasoning process here
The patient is a 35-year-old African American man with a recent history of anti-malaria prophylaxis prior to travel to Nigeria. He presents with fever, abdominal pain, weakness, jaundice, generalized pallor, and laboratory findings of bite cells and Heinz bodies on peripheral smear.

Key points:
- Bite cells and Heinz bodies are classic findings in G6PD deficiency, especially when triggered by drugs or infections (such as malaria).
- The patient has normal G6PD enzyme levels in the lab âĢĶ this is critical. The question states that laboratory tests show normal G6PD levels. However, in clinical practice, âĢľnormal glucose-6-phosphate dehydrogenase (G6PD) levelsâĢĿ may be misinterpreted âĢĶ but in reality, G6PD deficiency is a functional deficiency, and testing is typically for activity, not levels. The statement says âĢľnormal levels,âĢĿ which may suggest the patient does not have G6PD deficiency, yet the peripheral smear shows bite cells and Heinz bodies âĢĶ a hallmark of hemolysis due to oxidative stress.
- Anti-malarial drugs such as primaquine can cause hemolytic anemia in patients with G6PD deficiency. However, the patient is reported to have âĢľnormalâĢĿ G6PD levels âĢĶ so paradoxically, if it were G6PD deficiency, this would contradict the test result.

But here's the twist: The presence of bite cells and Heinz bodies suggests an oxidative hemolysis. While classical G6PD deficiency shows these features, the statement says âĢľnormal G6PD levels,âĢĿ which would suggest the G6PD enzyme is functioning normally.

This is a diagnostic trick: the patient presents with signs of hemolysis with oxidative damage and the most common cause of such findings in a traveler with malaria exposure and recent anti-malarial use is **G6PD deficiency**, even if enzyme activity isnâĢĻt tested correctly. However, given the explicit statement that the lab shows "normal glucose-6-phosphate dehydrogenase (G6PD) levels", this implies that there is no biochemical deficiency of G6PD.

Thus, in the absence of functional G6PD deficiency, and with the appearance of bite cells and Heinz bodies, another condition must be considered. The alternative is a drug or infection-related hemolysis. Since the patient had anti-malarial prophylaxis before going to Nigeria, and such drugs may cause hemolysis in rare circumstances in non-deficient individuals, however, this is still unusual.

But in Africa, autoimmune hemolytic anemia (AIHA), microangiopathic hemolytic anemia, or PNH may be considered. However:
- AIHA and PNH are less likely with only recent travel and no mention of autoantibodies or schistocytes.
- The findings of bite cells and Heinz bodies are typically seen in **oxidative hemolysis**, which is the hallmark of G6PD deficiency, even if lab testing is described as normal.

Wait âĢĶ the statement says âĢľlaboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels.âĢĿ That phrasing is odd because G6PD deficiency is not usually quantified by "levels" of protein âĢĶ rather, it's the activity (enzyme activity) that is assessed. So, "normal levels" could mean the lab test shows normal activity. This suggests that the enzyme activity is normal âĢĶ meaning, no G6PD deficiency.

Therefore, the presence of bite cells and Heinz bodies in the absence of G6PD deficiency is atypical. Bite cells are not typical of autoimmune or microangiopathic hemolysis. Heinz bodies, especially when extensive and associated with oxidative damage, are typical of G6PD deficiency.

Thus, the most likely cause, despite a "normal" G6PD test, must be misreading the clinical scenario. The phrasing "normal levels" may be misleading.